

Problem 2 (25 points). Here are some mathematical expressions in Outuan Hmong, all of which have only a single result:

- (1) $\text{pei}^{33} \text{pai}^{55} \text{au}^{33} \text{t}^{\text{h}}\text{u}^{13} \text{sau}^{33} \div \text{t}^{\text{h}}\text{u}^{13} \text{i}^{13} = 18$
- (2) $\text{pei}^{33} \text{pai}^{55} \text{qei}^{43} - (\text{t}^{\text{h}}\text{u}^{13} \text{tsi}^{33} \times \text{t}^{\text{h}}\text{u}^{13} \text{t}^{\text{u}}^{55})$
- (3) $\text{ɛoŋ}^{52} \text{t}^{\text{h}}\text{u}^{13} \text{qei}^{43} + \text{i}^{13} \text{t}^{\text{h}}\text{u}^{13} \text{qei}^{43}$
- (4) $\text{au}^{33} \text{pai}^{55} \text{ɛoŋ}^{52} \text{t}^{\text{h}}\text{u}^{13} \text{tsi}^{33} + \text{sau}^{33} \text{pai}^{55} \text{qei}^{43} - \text{t}^{\text{u}}^{55} \text{pai}^{55} \text{ljen}^{31} \text{t}^{\text{ou}}^{31}$
- (5) $\text{au}^{33} \text{t}^{\text{h}}\text{u}^{13} - \text{t}^{\text{h}}\text{u}^{13} \text{qei}^{43}$
- (6) $\text{tsi}^{33} \text{pai}^{55} \text{ljen}^{31} \text{qei}^{43} \div \text{pei}^{33} - \text{sau}^{33} \text{t}^{\text{h}}\text{u}^{13} \text{qei}^{43}$

Here are the results of the expressions above in random order (the result 18 is already given):

9, 18, 70, 76, 126, 152

(a) Determine the correct correspondences.

The following expressions all equate to the same number (some of them, however, can produce multiple results).

- (7) $\text{i}^{13} \text{pai}^{55} \text{ljen}^{31} \text{t}^{\text{u}}^{55} + \text{i}^{13}$
- (8) $\text{i}^{13} \text{pai}^{55} \text{sau}^{33} \text{t}^{\text{h}}\text{u}^{13} \text{t}^{\text{ou}}^{31} - \text{sau}^{33} \text{t}^{\text{h}}\text{u}^{13} \text{au}^{33}$
- (9) $\text{t}^{\text{ou}}^{31} \text{t}^{\text{h}}\text{u}^{13} \text{au}^{33} + (\text{i}^{13} \text{pai}^{55} \text{ljen}^{31} \text{tsi}^{33} \div \text{ɛoŋ}^{52})$
- (10) $\text{t}^{\text{h}}\text{u}^{13} \text{t}^{\text{ou}}^{31} + \text{i}^{13} \text{t}^{\text{h}}\text{u}^{13} \text{i}^{13}$
- (11) $\text{i}^{13} \text{pai}^{55} \text{tsi}^{33} \text{t}^{\text{h}}\text{u}^{13} \text{t}^{\text{u}}^{55} \div \text{i}^{13}$

(b) What is the number that all expressions (7-11) equate to?

(c) Write expressions (7-11) with digits in all possible ways.

(d) Write in digits:

- a. tsi^{33}
- b. $\text{au}^{33} \text{t}^{\text{h}}\text{u}^{13}$
- c. $\text{t}^{\text{ou}}^{31} \text{pai}^{55} \text{qei}^{43}$
- d. $\text{t}^{\text{u}}^{55} \text{pai}^{55} \text{ljen}^{31} \text{i}^{13}$

(e) Write in Outuan Hmong: 1, 17, 96, 102, 540.

△ Outuan Hmong is a variety belonging to the Eastern dialects of Hmu, a Hmong-Mien language. It is spoken by around 250,000 people in Hunan, China.

—Pan Tung-Le